

# Decision Theory

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# Roadmap

## From classical choice to embedded counterfactuals

- **Why decision theory matters for AI safety** (modelling, specification, reflective stability, multi-agent conflict)
- **Why it is hard:** optimisation is solved for *dualistic* agents but not for *embedded* ones, where the agent must *construct* counterfactuals
- **The candidates:** causal (CDT), evidential (EDT), functional/logical (FDT), updateless (UDT 1.0, UDT 1.1) – each is a different recipe for the counterfactual
- **A unifying lens:** CDT vs EDT as two readings of an action in sequential (AIXI-style) prediction – an *intervention* (chronological semimeasure) vs *evidence* (joint prediction)
- **The hard core:** problems with UDT, the 5-and-10 problem, logical counterfactuals and Löb's theorem
- **A unifying goal:** formalise decision problems as programs and ask which decision theory is *optimal* across all “fair” problems

# Why study decision theory? (for AI safety)

## Four reasons

- **Architecture-independent modelling.** We want to reason about how a superintelligent agent will *behave* without knowing the details of how it is built. Decision theory is a minimal model of “what is rational choice”, capturing behaviour any sufficiently capable agent tends to converge on.
- **Implementation / the specification–optimisation split.** Alignment decomposes into (1) *specify* a goal we want, and (2) *optimise* it correctly. Decision theory is part (2). A perfect specification is **not sufficient** for safety: if the agent constructs the wrong counterfactuals, it can “optimise” the goal in a way we never intended.
- **Reflective stability.** Safety properties must remain invariant under self-modification (cf. Agent Foundations deck). Decision theory is the simplest setting to ask: *would an agent running decision theory A want to self-modify into, or build a successor with, decision theory B?* A decision theory that wants to overwrite itself is a warning sign.
- **Multi-agent dynamics.** Decision problems get hard under strategic interaction and mutual modelling. We want a decision theory under which agents can *cooperate* with other superintelligences and *avoid catastrophic conflict*, without being *exploitable*.

# The easy case: dualistic agents

## When the agent is cleanly separated from the world

- A *dualistic* agent has clean input/output channels with its environment (the “cybernetic” picture: a player optimising a video game). It is “**larger than**” the world (can hold a full model of it) and “**outside**” it (need not reason about itself).
- Here the action is a genuine *free variable*: nothing in the world depends on the agent’s internal computation except through the downstream consequences of the action. Choosing optimally is conceptually solved:

$$a^* = \arg \max_a \mathbb{E}[U \mid \text{do}(A = a)].$$

- This is the mature, successful framework of **expected-utility maximisation**: the von Neumann–Morgenstern / Savage axioms (preferences satisfying coherence are EU-maximisation), Bayesian updating, and **complete-class theorems** (every admissible decision rule is Bayes-optimal for some prior).
- It underpins modern economics, statistics, and reinforcement learning. The conceptual work is done; only computation remains.

# The hard case: embedded agents

## When the agent is a part of the world it reasons about

- An *embedded* agent is a piece of the world it is trying to optimise (cf. Agent Foundations deck). It can be copied, predicted, simulated; its source code can be read by others.
- Now **“which action I take” is just another fact about the world**, fixed by physics (or by the agent’s own deterministic computation). There is no free variable to set, and **no ready-made answer to “what would happen if I acted differently”** – the alternative may be physically or logically impossible.
- So the counterfactual must be *constructed*, not read off. **This is the whole difficulty**: decision theory for embedded agents is the problem of constructing counterfactuals.
- Each decision theory is a different construction:
  - **EDT**: condition on the action as *evidence*,  $\mathbb{E}[U \mid A = a]$
  - **CDT**: *intervene* causally on the action,  $\mathbb{E}[U \mid \text{do}(A = a)]$
  - **FDT/UDT**: reason about the *logical output* of the decision procedure being  $a$

# Evidential decision theory: condition, don't intervene

## EDT as conditioning

- EDT evaluates an action by the *news* it would bring – it **conditions** on “I do  $a$ ” as evidence:

$$a^* = \arg \max_a \mathbb{E}[U \mid A = a].$$

- Conditioning treats the action as evidence about *everything* correlated with it – including facts the action does not cause.
- (We will see this *help* on Newcomb but *mislead* on the smoking lesion – once both rules are on the table.)

# A 60-second primer: Bayesian networks

## Encoding “what depends on what”

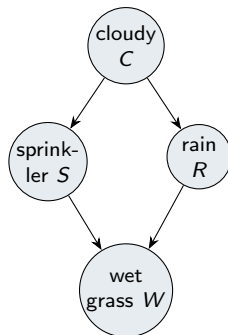
- A Bayesian network is a directed acyclic graph: nodes are random variables, an edge  $X \rightarrow Y$  means  $Y$  *directly listens to*  $X$ . The joint distribution *factorises* along the graph:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)),$$

with  $\text{Pa}(X_i)$  the parents (direct causes) of  $X_i$ .

- Each variable is independent of its non-descendants given its parents: a compact, structured stand-in for a huge joint distribution.
- For the graph on the right:

$$P(C) P(S \mid C) P(R \mid C) P(W \mid S, R).$$



*cloud cover is a common cause of sprinkler use  
and rain*

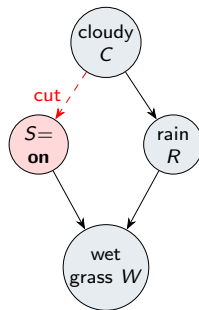
# Causal decision theory: seeing vs. doing

## The do-operator

- Pearl's key distinction: *observing*  $X=x$  (passive evidence, as in EDT) is not the same as *doing*  $X=x$  (an intervention).
- $\text{do}(X = x)$ : **sever all arrows into**  $X$  (cut  $X$  off from its causes), fix  $X = x$ , propagate downstream. Adjusting over the direct causes  $z = \text{Pa}(X)$ :

$$P(Y \mid \text{do}(X=x)) = \sum_z P(Y \mid x, z) P(z).$$

- Sprinkler graph: *seeing* the sprinkler on raises the chance it was cloudy, hence lowers the chance of rain. *Doing*  $\text{do}(S=\text{on})$  cuts  $C \rightarrow S$ , so it tells you **nothing** about rain.
- CDT**:  $a^* = \arg \max_a \mathbb{E}[U \mid \text{do}(A = a)]$  – judge each action by the causal consequences of *intervening* on it, holding the rest of the past fixed.
- The contrast in one line:  $\mathbb{E}[U \mid \text{do}(A=a)]$  (CDT, severs the action's causes) vs.  $\mathbb{E}[U \mid A=a]$  (EDT, keeps them as evidence).



$\text{do}(S=\text{on})$  severs  $S$  from its cause  $C$



# Prerequisites: the sequential prediction setting

## Histories, environments as chronological semimeasures, and beliefs

- Interaction: at each step  $t$  the agent emits an action  $a_t \in \mathcal{A}$  and receives a percept  $e_t \in \mathcal{E}$  (e.g.  $e_t = (o_t, r_t)$ ). The history is  $h_{<t} = a_1 e_1 \cdots a_{t-1} e_{t-1}$ .
- An **environment** is a **chronological semimeasure**  $\rho$ : it gives the probability of percepts *given* actions,

$$\rho(e_{1:t} \parallel a_{1:t}) \in [0, 1],$$

where the double bar  $\parallel$  marks the actions as **conditioned-on inputs, never predicted**.

- *chronological*:  $e_{1:t}$  may depend on  $a_{1:t}$  but not on *future* actions;
- *semimeasure*:  $\sum_{e_t} \rho(e_{1:t} \parallel a_{1:t}) \leq \rho(e_{<t} \parallel a_{<t})$  (probability may leak; a measure has equality).
- The agent's belief is a **Bayesian mixture** over a class  $\mathcal{M}$  of environments,  $\xi(e_{<t} \parallel a_{<t}) = \sum_{\nu \in \mathcal{M}} w_\nu \nu(e_{<t} \parallel a_{<t})$  (AIXI:  $\mathcal{M}$  = lower-semicomputable  $\nu$ ,  $w_\nu = 2^{-K(\nu)}$ ).
- Decision-theoretic crux: **when the agent chooses  $a_t$ , does that choice update its belief about which environment it is in?** The answer separates CDT from EDT.

# Chronological semimeasure = CDT

## The action is an intervention, not evidence

- In a chronological semimeasure the action appears **only to the right of**  $\parallel$ :  $\rho$  assigns *no* probability to actions, so “the action as evidence about  $\rho$ ” is not even definable.
- The posterior over environments updates on **percepts only**, holding the actions fixed as given:

$$w_\nu(h_{<t}) = w_\nu \frac{\nu(e_{<t} \parallel a_{<t})}{\xi(e_{<t} \parallel a_{<t})}.$$

Choosing  $a_t$  does **not** reweight  $w_\nu$ : *you do not update “which environment am I in” by deciding.*

- So the agent evaluates an action by *intervening* and propagating through a *fixed* which-world belief:

$$a_t = \arg \max_a \sum_{e_t} \xi(e_t \parallel h_{<t} a) [r_t + \dots] = \arg \max_a \mathbb{E}_\xi [U \parallel \text{do}(a)].$$

- This is exactly **CDT**: the chronological bar  $\parallel$  is the do-operator, lifted to whole histories. (Newcomb: such an agent two-boxes – the prediction is a fixed past percept, untouched by  $\text{do}(a)$ .)

# Joint prediction = EDT

## Actions predicted exactly like observations

- Now model the *entire* history  $h = a_1 e_1 a_2 e_2 \dots$  as **one stream to predict** over  $(\mathcal{A} \times \mathcal{E})^*$  – actions get probabilities too. Take the generative model to be a mixture over (world  $\nu$ , policy  $\pi$ ) pairs:

$$P(h_{1:t}) = \sum_{\nu, \pi} w_{\nu, \pi} \prod_{k=1}^t \pi(a_k | h_{<k}) \nu(e_k | h_{<k} a_k).$$

(Solomonoff's  $M(h) = \sum_{p: U(p)=h^*} 2^{-\ell(p)}$  is the universal such joint predictor.)

- Choosing/observing  $a_t$  is now ordinary **evidence**: Bayes updates the joint posterior over *both* the policy and the world,

$$w_{\nu, \pi}(h_{<t} a_t) \propto w_{\nu, \pi} \left[ \prod_{k < t} \dots \right] \pi(a_t | h_{<t}).$$

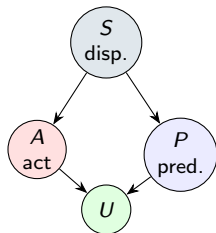
The factor  $\pi(a_t | h_{<t})$  tells you “**what policy am I running**”; and any world  $\nu$  *correlated with the action in the prior*  $w_{\nu, \pi}$  is reweighted – “**what world am I in**”.

- So the agent evaluates an action by *conditioning*:  $a_t = \arg \max_a \mathbb{E}_P[U | A_t=a, h_{<t}]$ .
- This is exactly **EDT** – and it reproduces the smoking lesion: the prior correlates the lesion (a world-feature) with the smoking disposition (a policy-feature), so conditioning on “I smoke” upweights lesion-worlds and the act looks like bad news, though it is no lever.

## Newcomb's problem: CDT two-boxes, EDT one-boxes

### The same problem splits the two rules

- **Newcomb:** a transparent box holds \$1,000; an opaque box holds \$1,000,000 *iff* a highly reliable predictor  $\Omega$  foresaw you would take *only* the opaque box.  $\Omega$  has already chosen.
- **CDT** applies  $\text{do}(A)$ : this **severs** your action from your disposition  $S$ , so it no longer informs  $\Omega$ 's prediction  $P$ . The contents are fixed; taking both boxes earns \$1,000 more *in every fixed state*. CDT **two-boxes** and predictably walks away with \$1,000.
- **EDT** conditions: one-boxing is strong *evidence* the opaque box is full, so  $\mathbb{E}[U \mid \text{one-box}]$  is huge. EDT **one-boxes** and gets \$1,000,000 – it keeps exactly the predictive dependence CDT discards.
- **Verdict:** CDT loses Newcomb. (In the lens above: the chronological-semimeasure agent two-boxes; the joint-prediction agent one-boxes.)



*CDT cuts  $S \rightarrow A$ , losing the link to  $P$ .*

# EDT mistakes evidence for control

## The smoking lesion

- A genetic lesion  $L$  causes cancer (cost  $C$ ) *and* makes you more likely to want to smoke. Smoking itself is harmless and gives benefit  $b > 0$ .
- Conditioning on “I smoke” raises  $P(L)$ , hence raises estimated cancer risk:  
 $\mathbb{E}[U \mid \text{smoke}] = b - C P(L \mid \text{smoke})$ . For strong correlation, EDT **abstains** – even though abstaining does *nothing* to reduce cancer. EDT confuses a **symptom** for a **lever**.
- **Scorecard:** CDT and EDT are each half right.
  - CDT ignores predictive dependence  $\Rightarrow$  loses Newcomb.
  - EDT overreacts to mere correlation  $\Rightarrow$  loses the smoking lesion.
- The missing ingredient is a dependence that is neither raw physical causation nor raw evidential correlation.

## CDT vs. physics: the intervention is a fiction

### Does CDT really “reflect physical reality”? (following Ebtekar & Hutter)

- CDT's  $\text{do}(A = a)$  holds the past fixed, **magically sets the action**, and propagates forward. What licenses cutting the action loose from its causes?
- In a *deterministic, time-reversible* physical world, your action has physical antecedents (your brain state, your source code). A predictor that reads those antecedents is correlated with your action through **perfectly ordinary physics**, not magic.
- “Severing the incoming arrows to  $A$ ” directly **contradicts embeddedness**: it pretends your action has no causes, but physically it does. The intervention is a harmless idealisation for a dualistic agent; for an embedded agent it *throws away real correlations* (with predictors, copies, simulations).
- So the usual defence is backwards: physics gives no privileged metaphysical “intervention” that frees the action from its past. This pressures us to move the counterfactual off the *physical act* and onto the agent's *policy / computation*.

# Spurious counterfactuals: EDT breaks for self-aware agents

## Conditioning on a probability-zero action

- A capable embedded agent can reason about (even *prove* things about) its own source code, so it can largely predict its own action. But then, for the action  $a$  it will *not* take,  $P(A = a) = 0$ .
- EDT's recipe  $\mathbb{E}[U \mid A = a]$  is then  $\frac{0}{0}$  – **undefined**. You may assign the unchosen action *any* value you like: a **spurious counterfactual**.
- So conditioning gives *no constraint* on counterfactual actions you are certain you won't take – exactly the cases decision-making turns on.
- (Foreshadow: in proof-theoretic form this same pathology becomes the **5-and-10 problem**, and the fix needs a notion of *logical* dependence.)

# Functional decision theory: choose your function's output

“Which output of this very function would yield the best outcome?”

- Reframe *what you are choosing*. You are not just *this body about to act*; you are running a fixed **decision function**, and your action is its *output* on the situation you face.
- Many things in the world may depend on that *same* function: a predictor who **modelled** it, a **copy** of you running it, a simulation of it. When the function's output changes, all of them change *with* it.
- So FDT does not ask “what should I do, holding everything else fixed?” (that is CDT). It asks the question the paper puts at its centre:

*“which output of this very decision function – given everything in the world that depends on it – would yield the best outcome?”*

and then *returns that output*. Concretely: score each candidate action *a* by the outcome in the world where *my decision function outputs a* (so the predictor predicted *a*, and every copy outputs *a*), and take the best.

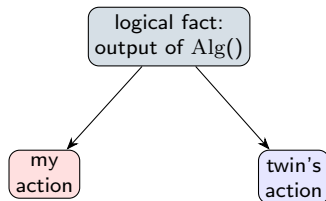
- It acts *as if it controls all instances of the computation at once* – not by magic, but because they really are the same logical fact.
- This wins the earlier cases for the *right* reason: in Newcomb the prediction tracks the function's output, so FDT **one-boxes**; the lesion does *not* run the function, so FDT still **smokes**. FDT shines exactly where embedded agents live – worlds containing copies of you, or predictors that model your decision procedure.



# Twin Prisoner's Dilemma & subjunctive dependence

## Same function, two bodies

- You and a psychological *twin* (your exact copy) play a one-shot Prisoner's Dilemma in separate rooms, no communication. You run the *same* decision function.
- **CDT**: "I can't causally affect her; defection dominates." Both defect, each gets \$1.
- **FDT**: "We compute the same function. If it outputs *C*, both cooperate; if *D*, both defect." So FDT cooperates – both get \$3.
- **Subjunctive dependence** (Y&S): when two physical systems compute the *same* function, their outputs are logically tied – like two calculators evaluating  $6288+1048$ : see one print 7336 and you know the other will too. Causal dependence is a *special case*; mere correlation (both calculators happen to be pink) is not. FDT acts as if it controls *all* instances of the computation at once.



*one logical cause, two  
physical effects: CDT  
sees no arrow be-  
tween them, FDT  
sees the shared parent*

# Counterfactual mugging: the updateless move

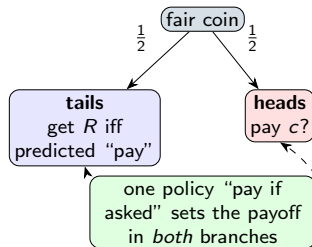
## Paying in a branch you know you're not rewarded in

- $\Omega$  flips a fair coin. **Tails:** you get  $R = \$10,000$  iff  $\Omega$  predicted you'd pay on heads. **Heads:**  $\Omega$  asks for  $c = \$100$ . You are asked to pay (so: heads). Pay?
- **Updateful** reasoning conditions on "I'm in heads": it sees only the \$100 cost and **refuses**. But *ex ante* the policy "pay if asked" is worth  $\frac{1}{2}(R - c) > 0$ .
- **UDT 1.0:** evaluate the local output against the **prior** – keep the (tails) branch whose payoff your policy controls, by dropping the updateful rule's extra  $O=o$  conditioning:

$$\text{choice}(o) = \arg \max_a \mathbb{E}[U \mid \text{choice}(o)=a].$$

So UDT 1.0 **pays**.

- Updatelessness is *not* ignorance (no predictor  $\Rightarrow$  UDT = ordinary EU max), and it is *reflectively stable*: the agent already follows the policy it would have precommitted to.



# UDT 1.1: optimise the whole policy, not one action

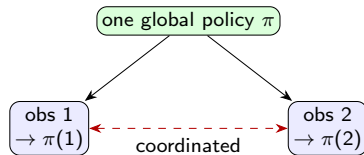
## When local outputs must be coordinated

- Even *updateless local-action* selection can be too local.  
**Copied-agent anti-coordination:** two copies see index 1 or 2, must pick *different* actions to win (10 each), else 0.
- The local question “if  $\text{Alg}(1) = A$ , what is  $\text{Alg}(2)$ ?” is the *wrong type*: both symmetric answers fail. The good object is the *whole table*  $\text{Alg} : \{1, 2\} \rightarrow \{A, B\}$ .
- **UDT 1.1 (Wei Dai):** first choose the entire *policy* (a map  $S : O \rightarrow A$ ) against the prior, *then* apply  $S^*$  to your actual observation  $o$ :

$$S^* \in \arg \max_{S: O \rightarrow A} \mathbb{E}[U \mid \text{choice}=S].$$

(UDT 1.0 instead fixes a single *output* and hopes the rest fall out right.)

- $\pi_{AB}, \pi_{BA}$  score 10; a shared tie-breaker coordinates the copies. This is **global optimisation over branches**, not local optimisation.



## The three shifts in “what am I choosing?”

CDT/EDT → FDT → UDT 1.0 → UDT 1.1 (after Kosoy’s formalism)

| Rule      | Object chosen           | Update status     | Dependence used        |
|-----------|-------------------------|-------------------|------------------------|
| CDT       | local physical act      | updateful         | causal intervention    |
| EDT       | local act (as evidence) | updateful         | evidential correlation |
| FDT / TDT | local algorithm output  | updateful         | logical / subjunctive  |
| UDT 1.0   | local algorithm output  | <b>updateless</b> | logical / subjunctive  |
| UDT 1.1   | <b>whole policy</b>     | updateless        | logical over policies  |

- **Shift 1 (FDT):** from a physical act to the output of an abstract computation – captures predictors, copies, simulations.
- **Shift 2 (UDT 1.0):** from posterior to prior – keep the value of being predictable, don’t discard branches whose payoff your policy controls.
- **Shift 3 (UDT 1.1):** from a single output to a whole policy – coordinate behaviour across all possible observations.

## Problem 1: paying for worlds that don't exist

### Ex-ante optimality is only as good as the prior

- UDT maximises *prior* expected utility. In the branch it actually inhabits it will **knowingly burn real utility**. In counterfactual mugging, after seeing heads, UDT hands over a real \$100 in the only world it will ever experience, to benefit a tails-world *it now knows did not happen*.
- Generalise: a UDT agent will take actions that are bad *in the actual world* because they would have been good in counterfactual scenarios that are never realised.
- This is dangerous when the prior is even slightly wrong about *which worlds are possible*: put prior mass on a predictor that does not actually exist (or on a logically impossible scenario), and UDT pays a **real, ongoing tax** for fake benefits. A clever (or merely hypothetical)  $\Omega$  can extract resources by promising/threatening in branches that never occur.
- The worry: is “best from the prior” the right notion of rational, when “the prior” ranges over worlds you have since learned are not actual?

## Problem 2: logical updatelessness

### Which prior, when some of your uncertainty is mathematical?

- A real embedded agent's uncertainty is not only *empirical* (coin flips). It is also **logical**: it does not know all consequences of its own source code, the  $10^{100}$ -th digit of  $\pi$ , or whether the opponent's program halts.
- UDT says "don't update." Refusing to update on *empirical* observations is clear. Refusing to update on *logical* facts is not: as the agent thinks longer it *learns* logical facts (proves theorems, computes digits) – and that looks exactly like updating.
- **Logical updatelessness** would have the agent choose its policy as if from a state of *logical* ignorance, not taking computed logical facts into account. But (a) it cannot help computing some of them, and (b) we have **no coherent "logical prior"** – a probability distribution over mathematical truths before any computation.
- This is open. A satisfactory UDT needs a theory of *logical uncertainty* (cf. logical induction), which we do not yet have. It is also the doorway to the 5-and-10 problem.

# The 5-and-10 problem

## Two programs (Demski & Garrabrant)

Take \$5 or \$10; utility is the dollars taken, so take \$10. The agent has its own source code and the world's, and chooses by proof search.

$U()$ :

```
if A() = 10: return 10
if A() = 5: return 5
```

$A()$ :

```
search for a proof of
  [A()=5  $\rightarrow$  U()=x] and
  [A()=10  $\rightarrow$  U()=y],
  for x,y in {0,5,10}
if found, x>y: return 5
else: return 10
```

## Why it returns \$5

- The world returns whatever the agent returns, so \$10 genuinely beats \$5. Yet with enough search time A returns \$5.
- Let  $P$  be the sentence " $[A()=5 \Rightarrow U()=5]$  and  $[A()=10 \Rightarrow U()=0]$ ".
- **A proof of  $P$  makes  $P$  true.** If  $P$  is proved, A finds it ( $x=5 > y=0$ ) and returns \$5. Then  $A()=5$ , so the first conjunct holds; and  $A()=10$  is false, so the second holds *vacuously* (a false "if" implies anything). Hence  $P$ .
- So  $\Box P \rightarrow P$  is itself provable ( $\Box X$  means " $X$  is provable"). **Löb's theorem** ( $\Box(\Box P \rightarrow P) \rightarrow \Box P$ ) then makes  $P$  provable; the search finds it and returns \$5.
- **Spurious proof:** A takes \$5 because it proves that taking \$10 pays \$0, and that holds *only because* it takes \$5. The action not taken is provably false, and an "if" with a false premise carries no information, so ordinary counterfactuals over one's own action break down.

# Why we want a formal optimality criterion

## Beyond case-by-case scoring

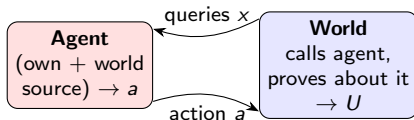
- Newcomb, counterfactual mugging, the 5-and-10 problem, the twin PD, open-source games all share a **common structure**: agent and environment are both *programs* that may inspect each other's code, simulate each other's behaviour, or contain subroutines algorithmically *isomorphic* to one another.
- Yet these are usually presented as standalone puzzles, and CDT/EDT/FDT/UDT are judged *case by case* ("does it get Newcomb right?") rather than against a single optimality standard.
- **The substantive question**: can we define one class of computational environments rich enough to contain all these problems at once, together with a criterion of optimality that any candidate decision theory can be *measured* against?



# Agents and worlds as programs

## The setup

- **Agent** = a program with access to (i) its own source code and (ii) the world's source code; it returns an action.
- **World** = a program that may
  - *call the agent* as a function (input  $\rightarrow$  action), possibly *many times on different inputs* (“what would the agent do if it observed  $x$ ?”), and
  - *search for proofs* about the agent's input-output behaviour;it returns a utility.
- **Goal:** design an agent whose utility is high across (almost) all worlds – the best mapping from situations to actions.



# Fairness

## Judging the agent only on its behaviour

- Without a constraint, a “world” could just punish source code: “if the agent is written in Python, utility = 0.” Then *no* decision theory could do well and the criterion would be vacuous.
- **Fairness:** the world may depend on the agent *only through its input–output behaviour* (its policy), never through implementation details unrelated to its outputs. Equivalently, a fair world is representable as a program that takes the agent’s *policy/behaviour* as input.
- Newcomb, the smoking lesion, counterfactual mugging, the twin PD, and the open-source PD are all **fair**: the predictor/lesion/copy interacts with the agent through its behaviour. This is what makes them legitimate tests.

# What a good agent must discover

## Two demands – which are exactly FDT/UDT turned into a spec

- **Coordinate across function calls.** If the world queries the agent on several inputs (probing its counterfactual behaviour), the agent must *recognise this* and keep its counterfactual outputs *coordinated* for high utility. (This is precisely UDT 1.1's move to optimise a whole policy.)
- **Subjunctive dependence.** If the world contains a subroutine *algorithmically isomorphic* to the agent (a copy in different code), the agent should treat its own decision as also fixing that subroutine's output – and so must *detect* that “the world contains something isomorphic to me.” (This is precisely FDT.)

## The open problem (and the bridge ahead)

- Define the class of fair computational worlds precisely; define optimality (e.g. dominance across all fair worlds); show CDT/EDT/FDT/UDT differ on it; and ask whether any (or some modification) is *optimal*.
- Caveats already met: logical counterfactuals (5-and-10), logical uncertainty / logical updatelessness, what counts as a “policy”, and multi-agent fairness & bargaining – which lead into reflective oracles, open-source games, and commitment races.